Karthik D K

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Summary

Firmware Engineer building the RISC-V software ecosystem for automotive and IoT at Bosch's TrusteD-V Innovation Team. 6+ years of embedded systems experience spanning kernel development, RTOS porting, and automotive ECU application software. Expertise in Rust, C, and Python for bare-metal and safety-critical systems. RISC-V Summit 2025 featured contributor.

Experience

Firmware Engineer | *Bosch Global Software Technologies* | *August 2019 - Present*

Hubris OS Port to RISC-V

- Ported Hubris OS to RISC-V - kernel primitives, trap/exception handlers, and PMP-based memory isolation for the Hazard3 core on Raspberry Pi Pico 2.
- Enabled dual-architecture support across ARM Cortex-M33 and RISC-V Hazard3 via custom linker scripts and startup code; worked around RP2350 errata and validated bring-up using OpenOCD/GDB over the Raspberry Pi Debug Probe.

TrusteD-V RTOS

- Developed proof-of-concept RTOS port for Mindgrove Secure IoT development board with working QEMU emulation support.
- Debugged kernel bring-up issues using Segger J-Link with OpenOCD and GDB.

Rust SDK for CDAC VEGA Processor

- Created the first Rust bare-metal ecosystem for the CDAC VEGA ARIES v3 board (THEJAS32 RISC-V SoC).
- Reverse-engineered register maps from the VEGA C SDK into a Python-based SVD generator, producing type-safe PACs for 15+ peripherals (UART, SPI, I2C, Timer, GPIO, ADC, PWM, PLIC) validated with bare-metal examples.

SiFive E34 Board Bring-up (Trina-Pi)

- Performed complete bare-metal bring-up of the UpBeatTech Trina-Pi board (SiFive E34 RISC-V core) entirely in Rust.
- Implemented a UART peripheral driver from scratch for serial console and debug I/O.

MLLib DecisionTree Inference

- Built a lightweight C DecisionTree inference engine for a resource-constrained PowerTrain ECU, using FlatBuffers for model serialization and calibration-tool parameterization.

Skills

Programming: Rust, C, Python

RTOS/Kernel: Linux, Hubris, xv6, bare-metal programming

Debug Tools: OpenOCD, GDB, tio, RP Debug Probe, Segger J-Link

Development Tools and Frameworks: WSL, Git, Claude Code, Buildroot, BusyBox, Cargo, Make

Open Source Projects

Contributions

- Active open-source contributor with merged pull requests into oreboot (Rust boot firmware) and the fastai and FluxML deep-learning libraries.

robot-hat-rs

- Developed Rust drivers (I2C, PWM, ADC) and a hardware abstraction layer for motor control, servo actuation, and analog sensor reading on the Robot HAT platform; published on crates.io with 4,000+ downloads.
- Powers picars, an autonomous PiCar-X vehicle with Rust-to-Python bindings for performance-critical control.

ixv

- Developed a Rust CLI for verifying Intel HEX files. Published on crates.io with 4,000+ downloads.

Technical Blog

- Writing on firmware, systems programming, and Rust - a public knowledge repository documenting my work.

Achievements

- **RISC-V Summit 2025:** TrusteD-V project contributor (featured talk).
- **Bosch Recognition:** Best Developer Award, Crowdsourcing Champ, AI Hackathon Winner.
- **Chess**: State-Level Championship participant.

Education

University Vishweshwaraya College of Engineering, Bengaluru — **75.2%**
B.E in Electronics and Communication — 2019

Certification — Machine Learning, Stanford University / Andrew Ng (Coursera, 2020) — Certificate